

Phishing Email Analysis Report

SOC Simulation Exercise

Analyst: Nandana K S Gowda

Analysis completed: March 2023

Report prepared: May 2026

Program: ANZ Cyber Security Management Virtual Experience - Forage

Emails reviewed: 7 (4 Malicious | 1 Spam | 2 Safe)

1. OBJECTIVE

This report documents the analysis of seven email samples to determine whether each was malicious, spam, or legitimate. Each email was reviewed for common phishing and social engineering indicators. A severity rating and recommended response action are included for each finding, consistent with how a Tier 1 SOC analyst would document a triage decision.

2. METHODOLOGY

Each email was reviewed independently using the following indicators:

- Sender address: does the domain match the claimed sender identity
- Subject line: does it use urgency, authority, or pressure tactics
- Links: are URLs obfuscated, mismatched, or pointing to unexpected domains
- Language and tone: grammar errors, unprofessional formatting, generic greetings
- Request type: does the email ask for credentials, account access, or personal information
- Context: is the email expected? Does it make sense given the apparent sender relationship?

Each email was assigned a verdict (Malicious, Spam, or Safe), a severity rating (High, Medium, Low, or None), and a recommended action.

3. ANALYSIS & FINDINGS

Detailed findings for each email are presented below.

Email 1 **SAFE** Severity: None

Subject: FYI

Sender: Adamm.johnnn1996@gmail.com (display: Adam John)

Analysis

This is a casual conversation between two friends discussing gaming trailers from Games Con. The reply from Velma Khan arrives within two minutes, which indicates a pre-existing relationship and that the message was expected. There are no suspicious links, no requests for information, and no unusual attachments. The informal language and context are consistent with personal communication between known contacts.

Recommended Action

No action required. This email is legitimate personal communication and can be classified as safe.

Email 2 **MALICIOUS** Severity: High

Subject: OneDrive Action Required

Sender: Venture.ru (claims to be Microsoft OneDrive)

Analysis

The sender domain is "Venture.ru", a Russian domain with no connection to Microsoft. The email claims to be from OneDrive and asks the recipient to update their Office 365 email address by clicking a link labelled "UPDATE YOUR ACCOUNT." The body contains multiple grammar and spelling errors ("recevie," "office 365" in lowercase, "Customers Support" in the sign-off). Legitimate Microsoft communications use official microsoft.com domains and maintain professional formatting throughout.

Recommended Action

Block and quarantine. Report sender domain to security team. Check email gateway logs to determine if any users clicked the link. If clicked, treat it as a potential credential compromise, force a password reset and escalate to Tier 2 for full account review.

Email 3 MALICIOUS *Severity: High*

Subject: *Is Facebook working for you?*

Sender: *Vinny (no surname, no verifiable email domain shown)*

Analysis

The email appears to be from a friend asking the recipient to test whether Facebook is working. The link provided <https://www.faceßbook.com.opt/login.htm> is not Facebook. The character "ß" in "faceßbook" is part of a homograph phishing technique that visually imitates facebook.com in an attempt to deceive users. This is a credential-harvesting phishing link designed to look like Facebook at first glance. It is also worth noting that the sender account itself could be compromised, which would explain why the email arrives from what appears to be a known contact.

Recommended Action

Block the link and quarantine the email. Alert the recipient immediately to not click. Check if the link was opened via email gateway logs. If the sender is a known colleague, notify them that their account may be compromised. Submit URL for sandbox analysis.

Email 4 SPAM *Severity: Low (Non-malicious)*

Subject: *Fwd: Drop + Koss GMR-54X-ISO Gaming Headset - Immersive 3D Sound*

Sender: *Drop (formerly Massdrop) via info@i.massdrop.com, forwarded by Adam Markus*

Analysis

This email is unsolicited marketing material from Drop/Massdrop, a legitimate audio and gaming peripheral retailer. The email has been forwarded twice, first from the mailing list to Adam Markus, then from Adam Markus to the recipient. There are no malicious links, no requests for credentials or personal information, and no suspicious attachments. The original sender domain (massdrop.com) is legitimate. This is classified as spam (unwanted commercial email) rather than a phishing or malware attempt.

Recommended Action

Mark as spam and unsubscribe from the mailing list if the recipient did not sign up for it. No security escalation required. Consider creating an inbox filter rule to automatically route similar marketing emails to the spam folder.

Email 5 MALICIOUS *Severity: High*

Subject: *You are needed*

Sender: *Vincent (no verified domain - temporary account claimed)*

Analysis

This email impersonates an FBI undercover agent and constructs an elaborate scenario to justify requesting access to the recipient's email account. It uses urgency ("before it's too late"), authority (FBI, classified investigation), and an unusual request for account access, all of which are classic social engineering techniques. No legitimate government agency or law enforcement body would contact a member of the public via a personal email account requesting access to their inbox for operational purposes. The lack of any verifiable sender identity, official domain, or contact details confirms this is fraudulent.

Recommended Action

Do not engage or reply. Block the sender. Report to the security team as a social engineering attempt. Notify the recipient that this is a scam. If the recipient provided any account access or responded with personal details, escalate immediately for account security review.

Email 6 **SAFE** Severity: None**Subject:** RE: WFH**Sender:** Reuben.corrigan@anz.com (Reuben Corrigan, Cyber Security Trainee, Group Technology ANZ)**Analysis**

This is a legitimate internal workplace email exchange between two ANZ employees, Reuben Corrigan and Alan Bryce, who are both members of the Group Technology ANZ team. Both sender and recipient email addresses use the official anz.com domain. The email signature matches the sender name and includes a verified ANZ address and office location. The content is a straightforward reply about a work from home arrangement, a project update, and a coffee catch-up, all of which are consistent with normal internal communication. There are no links, attachments, or requests for sensitive information.

Recommended Action

No action required. This email is legitimate personal communication and can be classified as safe.

Email 7 **MALICIOUS** Severity: High**Subject:** Did you know you can save up to 15% off your car insurance if you switch to Geico?**Sender:** Val.kill.ma (claims to be Mike Ferris)**Analysis**

There are three clear indicators of malice in this email. First, the display name of the sender is "Val.kill.ma", which does not match the signature name "Mike Ferris." Second, the email claims to be from Geico Insurance but the sender address has no connection to any official Geico domain. Third, the only link in the email body is obfuscated using "hxxp" notation (hxxp://iwhrhwic.y.urlif.y/receipt.php), which is a deliberate technique used to bypass automated link-scanning tools while still directing human readers to a potentially malicious URL. Legitimate insurance companies do not use obfuscated links or mismatched sender identities.

Recommended Action

Block the sender and quarantine the email. Submit the defanged URL to a sandboxed URL analysis platform such as VirusTotal or URLscan.io for further assessment. Report the sender domain to the threat intelligence team. Advise the recipient not to interact with the email. Monitor for similar emails targeting other users.

4. FINDINGS SUMMARY

The table below summarises the verdict, severity, and primary indicator for each email. Classification reflects final reviewed verdict after validation against model answers and post-analysis calibration.

No.	Verdict	Severity	Key Indicator
1	SAFE	None	Casual conversation between known contacts; no threat indicators
2	MALICIOUS	High	Russian sender domain impersonating Microsoft OneDrive
3	MALICIOUS	High	Homograph phishing URL mimicking facebook.com
4	SPAM	Low	Unsolicited forwarded marketing email from legitimate retailer
5	MALICIOUS	High	FBI impersonation requesting email account access
6	SAFE	None	Legitimate internal ANZ workplace email; verified domain
7	MALICIOUS	High	Sender/display name mismatch, obfuscated hxxp link, no Geico domain

5. KEY INDICATORS OBSERVED

Across the four malicious emails, the following phishing indicators appeared most frequently:

- Sender domain mismatch: claimed identity does not match the actual sending domain
- Urgency and authority tactics: language designed to pressure the recipient into acting quickly
- Homograph and lookalike URLs: domains crafted to appear legitimate at first glance
- Obfuscated links: hxxp notation used to bypass automated link scanning
- Poor grammar and unprofessional formatting: inconsistent with legitimate corporate communication
- Requests for account access or credentials: almost always a sign of malicious intent
- Display name / signature mismatch: sender name differs from the signature block

6. ANALYST REFLECTION

Note: This section was added during the preparation of this formatted report in May 2026, after reviewing the original analysis against the ANZ/Forage model answer.

Upon reviewing my initial findings, I identified two instances where I incorrectly classified benign activity as malicious. These have since been corrected in the report to better reflect an accurate SOC triage process.

Email 1 was originally flagged as malicious. On re-evaluation, the email is clearly a casual and legitimate conversation between known contacts. The quick reply timing, informal tone, and absence of suspicious links or attachments support a Safe classification.

Email 4 was originally classified as malicious with a Medium severity rating. However, the email is legitimate marketing content from a recognised retailer and is more accurately classified as Spam (Low Severity). Although unsolicited, it contains no credential harvesting attempts, malicious links, or social engineering indicators.

These corrections reflect my continued improvement in distinguishing between malicious activity and legitimate but unwanted content. This distinction is important in a real SOC environment, where incorrectly classifying safe or low-risk emails can waste analyst time and contribute to alert fatigue. Reviewing these mistakes helped me better understand the importance of analysing context, sender legitimacy, and user intent together rather than relying only on initial suspicion. It also reinforced how techniques such as homograph phishing and sender domain mismatches can be strong indicators of malicious intent when combined with suspicious behaviour.

7. CONCLUSION

Four of the seven emails were confirmed malicious, one was classified as spam, and two were safe. The malicious emails ranged from a straightforward credential-harvesting attempt using a Russian domain (Email 2) to a more sophisticated homograph phishing attack designed to look like Facebook (Email 3). The FBI impersonation in Email 5 used a multi-step social engineering narrative, the kind of attack that targets people emotionally rather than technically.

The two safe emails, a personal conversation (Email 1) and an internal workplace exchange (Email 6), both showed consistent sender identity, appropriate context, and no requests for sensitive information. These are the markers of legitimate communication.

Technical controls such as sender authentication (SPF, DKIM, DMARC), email gateway filtering, and link scanning can catch many of these threats. However, as Email 3 shows with its homograph URL and Email 5 shows with its narrative-driven manipulation, technical filters alone are not sufficient. User awareness and structured analytical thinking remain essential layers of defence.

DISCLAIMER

This report presents analysis originally completed as part of the ANZ Cyber Security Management Virtual Experience Program (Forage, March 2023). The email samples were provided by ANZ and Forage as part of a structured, self-paced virtual experience program for educational purposes. All findings, severity ratings, and recommendations are the original work of the analyst. No ANZ/Forage proprietary documents or original source files are reproduced in this document. This formatted document was prepared in May 2026 for inclusion in a professional cybersecurity portfolio.